

ASSIGNMENT: WIRELESS NETWORKING

Course: Computer Networks & Communications | Status: Pending / Completed Submission

Instructions: This document contains completed solutions and structured answers for all four tasks outlined in the Wireless Networking assignment. Formatted strictly in professional black-and-white print style.

Task 1: Popular Wi-Fi Standards & Key Differences

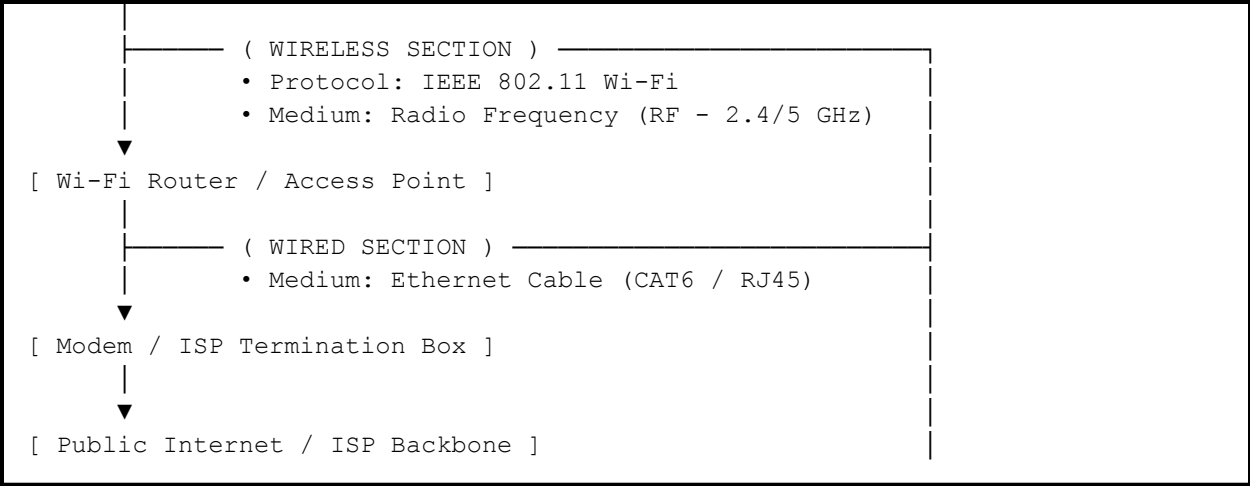
The table below details three popular IEEE 802.11 Wi-Fi standards, highlighting their respective frequency bands and maximum theoretical data rates / key performance differences.

Standard	Frequency Band(s)	Key Difference in Speed / Bandwidth
802.11n (Wi-Fi 4)	2.4 GHz and 5 GHz	Introduced MIMO (Multiple Input Multiple Output) technology, achieving speeds up to 600 Mbps.
802.11ac (Wi-Fi 5)	5 GHz only	Higher channel bandwidths (up to 160 MHz) and denser modulation (256-QAM), reaching multi-gigabit speeds up to 3.5 Gbps.
802.11ax (Wi-Fi 6)	2.4 GHz and 5 GHz (plus 6 GHz in 6E)	Uses OFDMA and MU-MIMO for extreme efficiency in crowded environments, delivering peak rates up to 9.6 Gbps.

Task 2: Home Network Topology Diagram

The schematic layout below illustrates the connection path from a smartphone to the external internet via a home Wi-Fi router, clearly demarcating the wireless and wired segments.

[SmartPhone]



Task 3: Wi-Fi Environment Scan & Selection Analysis

A local Wi-Fi scan captured the following parameters across three visible networks in the vicinity:

SSID (Network Name)	Signal Strength	Security Type	Band / Channel
HomeNet_5G	-45 dBm (Strong)	WPA3-Personal	5 GHz (Ch 36)
CoffeeShop_Guest	-67 dBm (Moderate)	Open / Captive Portal	2.4 GHz (Ch 6)
NeighborNet_2.4	-82 dBm (Weak)	WPA2-AES	2.4 GHz (Ch 11)

Network Selection & Justification: I would choose **HomeNet_5G**. This network offers the strongest signal strength (-45 dBm), ensuring high throughput and low packet loss, combined with state-of-the-art WPA3 encryption which provides robust protection against offline dictionary attacks and individual session hijacking. CoffeeShop_Guest lacks encryption entirely, and NeighborNet_2.4 suffers from a weak signal (-82 dBm) and older WPA2 protocol.

Task 4: SSID Hiding — Concept, Advantages, & Disadvantages

Definition of SSID Hiding: SSID (Service Set Identifier) hiding involves configuring a wireless router to stop broadcasting its network name in periodic IEEE 802.11 management frames known as beacon frames. When enabled, the network does not appear in standard device Wi-Fi scan lists, requiring users to manually input both the exact SSID name and security credentials to establish a connection.

1. Advantage of SSID Hiding:

- Reduces Casual Discovery: It prevents casual passersby, neighbors, or uninitiated visitors from seeing your network name pop up on their devices. This reduces opportunistic connection attempts and shields your router from random unauthorized probe requests or surface-level targeting.

2. Disadvantage of SSID Hiding:

- **False Sense of Security & Increased Client Overhead:** SSID hiding does not actually make a network invisible; the network name is still transmitted in plaintext whenever a connected device actively probes for it or reassociates. Furthermore, it causes client devices to continuously broadcast probe requests containing the hidden SSID name wherever they go, exposing the network name to malicious wireless sniffers. It also creates inconvenience when onboarding new devices.